Individual Assessment Coversheet

To be attached to the front of the assessment.

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature	Date
Bista	30 August 2024

Lecturer's Comments:

Marks Awarded:	%

Signature	Date

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Section A

Question 1

1.1) In order for something to be considered self-aware it needs to be able to observe and understand their own experiences, feelings and thoughts from the surrounding world and people and to be aware of how the consequences of their actions affects themselves and others. Self-awareness can encompass the ideas of self-perception, self-consciousness as well as self-reflection (Dell'Aversana, 2024).

Current AI systems are created after being trained through the deep learning method, which refers to a branch of AI that is built of a structure that imitates the human brain through numerous layers of neural networks. Deep learning allows these AI systems to train from data that can be non-linear or high-dimensional. This data can include images, text as well as speech and enables AI to perform complex tasks face recognition and translate languages. These AI systems are also paired with natural language processing enabling the AI to generate human-like responses to questions and create summaries as well as information analysis. Furthermore, these AI systems may also be given computer vison allowing it to analyse visual information and perform object detection as well as scene understanding.

Given the current state of AI development, AI can be self-aware in the sense that it can interpret its environment through computer vision as well as understand and provide human-like responses to information that it has been fed. However, the current state of AI development does not provide these AI systems the ability to experience human emotions and thoughts preventing it from undergoing self-perception, self-consciousness as well as self-reflection. Therefore, AI will not become fully self-aware given its current state of development.

1.2) The application of the ethical framework of AI can provide a safety net for organisations developing or integrating AI systems. The AI ethical framework can support individuals in these organisations to determine the most fundamental ethical principles that need to be taken into consideration to ensure that they develop or utilise AI systems in an ethical manner.

Moreover, this ethical framework focus on the ethical practices of AI systems in general, which enables it to be considered during the five stages of the AI lifecycle consisting of requirements engineering, design, implementation, verification and operation. Therefore, it can improve employees understanding of their AI system, its capabilities as well as its limitations (Wiesche, 2023). This improved understanding enables these employees to identify and perform essential, ethical decisions more effectively and efficiently.

1.3) Sapience refers to a state of awareness that allows for cognition abilities such as thinking, reasoning and decision-making (Yolles, 2022). AI systems with sapience capabilities can prove to be beneficial for humans as it can make predictions and choices based on large amounts of data that it has analysed.

Applications of this type of AI systems can be used in determining the appropriate diagnosis for patients after analysing medical information. This medical information can include images and can be utilised in various healthcare sectors including radiology, dermatology, pathology as well as cardiology (Bajwa, 2021). Furthermore, these AI systems can also be used to suggest or make informed decisions for financial companies after detecting patterns and trends in the financial industry.

1.4) A sapient AI system should not be utilised in business associated with the gambling industry like casinos as it violates the ethical component of non-maleficence. This ethical component can be violated by gambling establishments by being used to determine methods to increase customer engagement based on customer behaviour to determine. This can be harmful for their customers, because it also increases the likelihood of their customers to become addicted to gambling and put them into financial struggles due to the addiction.

Moreover, sapient AI systems should also not be used in the marketing industry as it violates the ethical component of privacy. Privacy can be violated as sapient AI systems can be used to market specific products and services after observing and analysing the behaviour of users. The user's data that is gathered can be done without their knowledge or consent, which defies this ethical component.

1.5) Given the current development of AI systems and models, it proves to be beneficial for humanity. The improvements towards the sapience within AI systems will increase overall productivity and efficiency when completing complex tasks due to its ability to easily analyse large volumes of data in various scenarios and industries. Additionally, sapience also delivers the advantage of the AI systems being able to determine accurate predictions based on any patterns it detects in the data, enabling for individuals to make informed choices. Moreover, the capabilities provided by AI systems allow for the automation of repetitive tasks enabling for a further increase in productivity as individuals can focus on more complex tasks.

Question 2

2.1) The three UN Sustainable Development Goals (SDG) that would benefit the most from AI integration is SDG 4, good health and well-being, SDG 13, climate change, and SDG 16, peace, justice and strong institutions.

2.2 a) In terms of efficiency and innovation for SDG 4, the implementation of AI systems in healthcare institutions has proven to beneficial due to its ability to automate simple tasks enabling healthcare professionals to complete tasks at quicker rate. AI systems are able to scan through and analyse large volumes of patient data more efficiently and accurately when compared to humans (Lau, 2023). In addition, AI systems can act as a catalyst for innovation within the healthcare industry by improving already existing equipment to be more effective and create new devices that fully utilises AI capabilities for patient care.

For SDG 13, AI systems are used to help optimise the consumption of energy in buildings and systems to save on limited natural resources required to produce power (Chaudhary, 2023). Additionally, AI systems can innovate how instances of climate change can be monitored and predicted due to its ability to analyse large data sets. This also increases the efficiency of government institutions as they can develop plans that will counteract these instances of climate change before it occurs.

For SDG 16, AI systems has innovated cybersecurity by providing systems that detects for patterns of potential crimes and flags it. This innovation also increases efficiency, because any threats detected will begin an investigation immediately where these threats can be halted in a quicker manner before any harm takes place. These AI systems further innovates crime detection by continuously improving its accuracy at predicting threats after it learns from data that it interacts with (Truby, 2020).

2.2 b) The applications of AI systems in SDG 4 can include utilising its ability to analyse patient data to determine the most appropriate diagnosis and treatment for patients. Furthermore, the capabilities associated with AI systems allows it to support patients in requiring motor functions and skills (Lau, 2023). AI systems can also be applied to general healthcare processes such as monitoring the vitals of patients and notifying staff if an abnormality occurs.

The integration of AI in SDG 13 can be seen by Japan, which uses an AI system to inspect the deforestation in the Amazon Rainforest as well as design urban areas to be more sustainable and less harmful to the environment. Moreover, it can also be implemented to improve the optimisation of both solar and wind energy over the electrical grid to help support the future transition from harmful power supplies, like coal-fired power stations, to renewable energy sources (Chaudhary, 2023).

An application of AI in SDG 16 was performed by anti-money laundering (AML) enforcement agencies that have created their own AI systems that were designed to detect and notify them of acts of money laundering more effectively than humans (Truby, 2020). In addition, financial institutions can also implement AI systems into their software for the purpose of detecting and prevent fraud and other financial crimes. AI systems can also be used to test the security of software programs by searching for any vulnerabilities that fixed after the test is complete.

2.2 c) The challenges associated with AI systems in SDG 4 includes the safety of the patient such as the possibility for an AI system to provide an incorrect diagnosis if it is trained on biased or inaccurate data. Instances of incorrect diagnoses from an AI system can also be produced if the patient is an unrepresented minority where there is a lack of clinical data (Lau, 2023). If any harm is done to patients due to inaccurate diagnoses, then the healthcare institution will be responsible for not ensuring that the patient receives the appropriate care.

A challenge of AI systems in SDG 13 is similar to one of the challenges of AI in SDG 4 where the AI system could be taught off of biased data that makes it more difficult for the AI system to accurately predict climate change. Furthermore, AI systems can also have a large carbon footprint due to the processing power required for it to analyse large amounts of data (Chaudhary, 2023). In terms of accountability, multiple companies from various industries that makes use of AI systems can contribute to these challenges.

A major challenge associated with AI systems in SDG 16 is the ability for criminals to utilise AI programs to easily commit crimes and avoid the detecting of other AI systems (Truby, 2020). Additionally, there is the possibility for AI systems designed to detect and flag threats to waste the time of investigators by flagging instances that were not threats. Moreover, accountability will lie on organisations who do not ensure that their security AI systems are constantly updated to protect against any recent developments in hacking technologies and techniques.

Question 3

3.1) The additional components for the ethical use of AI that are not in the current framework consists of contestability as well as reliability and safety. Contestability refers to ensuring that individuals are able to contest the use of AI systems that impacts the individual or their community. In order to make this ethical component effective is to provide individuals access to the algorithms of the AI and a system of oversight must be implemented. Reliability and safety refers to ensuring that AI systems reliably operates their intended purpose and is not a threat to the safety of individuals. These systems must be monitored to guarantee that the AI system continues to follow this ethical aspect (Australian Government, 2024).

3.2) The current AI ethical framework is sufficient to address various ethical challenges associated with AI systems. A scenario where an AI system is used to spread misinformation to exploit individuals and manipulate public opinion is an example of unethical AI usage (Capitol Technology University, 2023). This is addressed in the beneficence aspect of the framework that states that AI systems should be developed and be utilised for doing good.

Moreover, an AI system can use personal and private information from people and does not preserve the privacy and security of users. This is covered in nonmaleficence where AI systems should not inflict harm on others and the privacy of individuals is a subset of this component, since it can be used to harm the individual. Other AI systems can be designed to integrate personalisation that can’t be controlled the user, which can undermine the ability for individuals to make their own decisions as some choices will not be given due to the personalisation (Council of Europe, 2024). This issue is involved in the autonomy aspect that indicates that AI systems must respect the capacity for people to make their own choices.

In addition, many AI systems are trained off of large data sets and there are situations where the result produced by the AI system will be incorrect and unfair due to biases in the data (Capitol Technology University, 2023). This ethical issue is addressed by the justice component of the framework that indicates that AI systems individuals must face situations of bias and discrimination. Furthermore, the processes required for AI systems to produce outputs are opaque in a sense that individuals will not be aware of why the AI system produced the result due to complex code and decision making (Council of Europe, 2024). This problem is covered in the explicability aspect of the framework where AI systems need to be transparent on how it arrives to its decisions in a way that individuals can easily understand. This is most beneficial if the AI system is producing incorrect results and people want to understand where these issues occurred in order to fix it.

3.3) The most effective method to enforce an ethical framework is to make any violations of it punishable by law to reduce the likelihood of an individual or organisation from committing unethical actions. A scenario of this occurring is any instances of patient data being mishandled by AI-powered healthcare systems will be considered a HIPAA violation and the healthcare institution will be punished accordingly. To ensure that any ethical violations, during the deployment or development of AI systems, are discovered by the agencies that enforce this ethical framework, a reporting system should be implemented. A reporting system will allow individuals inside or outside organisations to report any instances of ethical violations to the agency where they can begin an investigation to determine if any aspects of the ethical framework have been violated. Moreover, these reporting systems must provide a secure and safe environment that protects individuals from the offending party, which will make them more likely to report ethical violations.

Question 4

AI systems that are trained over datasets can limit diversity and inclusivity due to biases within the datasets. The bias in AI systems refers to any unfair outcomes or discrimination due to errors that occurred during the decision-making process of the AI system. Bias in AI systems is an important issue when it comes to diversity and inclusivity, because there are many ways for AI to become biased in various industries when it makes decisions.

These sources of AI biases can occur at various stages of the AI pipeline such as the collection of data, how data is interpreted, how the algorithm for model training was designed and the how users interact with an AI system that is deployed (Chapman University, 2024). Each source of bias can arise during any of these processes of an AI system and are called data, algorithmic and user bias.

Data bias refers to biased outcomes that happen due to the quality of the data that the AI system is trained off of. Low quality data that can lead to data bias includes any data that does contain any relevant information, it is incomplete, or it was already biased due to it being collected from a biased source. Algorithmic bias refers to how the algorithm, used by the AI system to train off data, is biased. A biased algorithm can be created if it was built with biased assumptions or criteria during the decision-making process. User bias is the introduction of any biases and prejudices from users that interact with the AI system and can impact the system if the users provide data for the purposes of training the AI model (Ferrara, 2024).

These different sources of bias that can occur, indicates how easy an AI system can become biased and harm diverse groups of individuals at different magnitudes depending on how these AI systems are implemented. If an AI system with algorithmic bias is utilised in the hiring process of an organisation, then certain individuals could be overlooked by the AI system. This can occur depending on any bias that was implemented into the hiring criteria that the algorithm will search for in candidates and this design flaw that limits diversity and inclusivity could have been done intentionally or by accident.

Moreover, an AI system designed to diagnosis patients in a healthcare facility can be incorrect and harm the well-being of patients due to data bias. Data bias can happen in this scenario if the patient is from a diverse group where the AI system has limited medical information about and can determine an incorrect diagnosis and treatment plan that will harm the patient. In addition, an AI system designed to identify criminals can misidentify innocent individuals due to user bias. Instances like these can occur by users inputting biased information that makes it more difficult for the AI system to accurately identify a criminal if they are a member of a diverse group.

These biases can be solved or minimised by implementing solutions based on the components of the AI ethical framework. Beneficence can be accomplished by assigning an employee to oversee the output of an AI system. Doing so will allow the employee to identify any errors and misconceptions produced by the AI system, preventing the potential harm of diverse individuals and correcting the content of the output. Additionally, nonmaleficence should be taken into account when developing an AI system to ensure that the AI model will either be less likely to become biased or not biased at all. This can be done by identifying all potential vulnerabilities for biases in the structure of the algorithm for the AI system and ensuring that these vulnerabilities are solved before it is deployed.

Furthermore, autonomy can be achieved by incorporating a diverse workforce to observe the operations of the AI system and request configurations to be made to it when they determine any instances AI biases against diverse individuals and groups. In addition, justice is accomplished by preventing data bias from occurring by ensuring that the data is representative and monitor the state of the dataset with the use of a bias risk assessment tool (Judy Wawira Gichoya, 2023). Moreover, explicability can be introduced to solve instances of bias by making the process of decision-making from the AI system transparent so that employees are able to identify the source of the bias and create a solution to the issue.

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